



ISTE 430 – Information Requirements Modelling

Spring 2026

Group Project

Travel Ease

Group #5

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Manifesto

This project was completed collaboratively, with responsibilities distributed across team members based on different phases of system development.

Daniyal Zharov played a leading role throughout the project, contributing across all phases of the System Development Life Cycle. He conducted and managed stakeholder interviews for requirement gathering, developed all Data Flow Diagrams (Context, Level-0, and Level-1), and created decision tables to model system logic. Daniyal also designed the conceptual model and contributed significantly to the logical modeling phase by defining entities, attributes, relationships, and ensuring consistency across all system components. He further supported the later phases by contributing to the physical data modeling and participating in the implementation of the system, helping ensure that the database design, user interfaces, and application logic remained aligned with the original analysis and design decisions.

Shuja Ahmad contributed to **Phase 3(b): Logical Modeling**, assisting in refining the relational schema, identifying keys and relationships, and supporting normalization to Third Normal Form (3NF).

Fatmah was responsible for **Phase 3(c): Physical Data Modeling** and **Phase 4: Implementation**, including translating the logical model into a physical database schema, defining data types and constraints, and developing the Flask-based web application that connects the user interface to the MySQL database.

All team members participated in reviewing the project and providing feedback to ensure overall consistency and correctness.

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Phase 1 - Systems Planning and Selection

a. Need for the System

Planning a trip today often requires using multiple separate platforms for different tasks such as booking flights, reserving hotels, and checking visa requirements. These systems are not integrated, which makes the process inefficient and time-consuming. Users have to switch between platforms, compare information manually, and keep track of important details on their own. This increases the risk of missing critical information, especially when it comes to visa regulations or travel restrictions.

From a systems perspective, this highlights a clear gap between what users need and what current systems provide. Existing platforms are designed to handle specific services rather than the full travel experience, which results in a disconnected workflow. Instead of supporting users, the systems force them to manage the process themselves. TravelEase is designed to address this issue by bringing all key travel-related functions into a single platform, allowing users to plan their trips more efficiently and with less effort.

The system is centered around the user, who can create and manage trips by selecting destinations and organizing related bookings such as flights and accommodations within one place. Each trip acts as a structured plan that groups all relevant travel components, making it easier to keep track of information and avoid confusion. In addition, the system connects destination data with visa requirements and seasonal recommendations, helping users make more informed decisions.

The benefits of this system are both practical and experiential. On the practical side, it reduces planning time, minimizes errors, and improves access to accurate information. On the experiential side, it improves user confidence, simplifies decision-making, and creates a smoother and more organized planning process.

b. Project Scope

TravelEase is a web-based system that supports users in planning and organizing their travel through a single, integrated platform. The system allows users to search for and manage flights and hotel bookings, access visa-related information based on their destination, and explore travel options depending on the season. It also enables users to organize their trips in a structured way, where all related details are stored and managed together.

The system is intended for individual travelers who want a more efficient and organized way to plan their trips. It focuses on combining information and services from different sources into one interface, rather than replacing existing providers. For this reason, the system does not process visa applications, handle payments, or operate airlines and hotels. Instead, it acts as a centralized platform that supports planning and decision-making.

The boundaries of the system are limited to managing travel-related information and user interactions within the platform. It does not extend to external service execution or real-world operations. Defining the scope in this way keeps the system realistic and ensures that it can be properly developed within the constraints of the project, while also providing a strong foundation for further analysis and design in later phases.

Phase 2 - Systems Analysis

a. Requirement Determination

i. Interviewee Selection

In order to accurately determine the requirements for TravelEase, interviews were conducted with three different individuals, each with a different perspective regarding the system. Such a selection of interviewees will ensure that the system is understood in its entirety.

The first interviewee was an individual who travels frequently with the company. The interview with this individual was used to determine how the individual currently plans their trips, any difficulties in the interactions with various platforms, and what features they may expect from a platform like TravelEase.

The second interviewee was an individual that works for a travel agency and is responsible for booking travels for customers for their agency. Through interviewing this individual, the requirements regarding the information that must be obtained for bookings can be determined.

The third interviewee is an individual who manages the IT systems of the company. Through interviewing this individual, the requirements regarding the data that is to be stored by the system and how the system is to be limited can be determined.

These various interviewees will ensure that the requirements of the system reflect both the users of the system, as well as the way in which the system is to be structured.

The insights obtained from these interviews will help to create the process and data models for TravelEase.

ii. Interview Questions and Justification

For each interview, a combination of open-ended and closed-ended questions were asked of each interviewee.

Open-ended questions ask for an explanation or description from the interviewee of a certain topic. These types of questions are useful in the gathering of requirements in that they can help to uncover problems and requirements that are not yet specifically recognized by the interviewers. Furthermore, these types of questions allow for the interviewees to describe the current processes and systems within their industry.

Closed-ended questions ask for a specific answer to a question from the interviewee. These types of questions are useful in confirming certain requirements of a system. Their answers can help to confirm the features that are to be included in the system, as well as the limitations of that system.

By employing both types of questions, a determination of the requirements for the system can be made. The open-ended questions will reveal any new requirements that have been overlooked by the developers. In contrast, the closed-ended questions will help to confirm the features and limitations of the system. Thus, both types of questions will ensure that the requirements for the system are both complete and actionable.

The table below presents the questions that were asked of each interviewee, along with the justification for each question type:

Interview	Question	Type	Justification
Traveler	How do you currently plan your trips?	Open-ended	Allows the user to describe the full planning process, helping identify key activities that will later be modeled in system processes (DFDs).
Traveler	What problems do you face during trip planning?	Open-ended	Reveals user pain points and inefficiencies that the system must address.
Traveler	How much time do you typically spend planning a trip?	Open-ended	Provides measurable insight into process inefficiency, supporting the need for system improvement.
Traveler	Do you ever miss important information such as visa requirements?	Closed-ended	Confirms the presence of critical risks in the current process, justifying inclusion of visa-related features.
Traveler	Would you prefer a system that combines all travel services in one place?	Closed-ended	Validates the core concept and value proposition of the proposed system.
Booking Staff	What information is required when organizing a trip for a customer?	Open-ended	Helps identify essential data elements that will later form system entities in the ERD.
Booking Staff	What are the most common issues you encounter during the booking process?	Open-ended	Identifies process-level inefficiencies that will be addressed in DFD modeling.
Booking Staff	How do you currently manage customer bookings across different platforms?	Open-ended	Highlights integration challenges and supports the need for a centralized system.
Booking Staff	Would a centralized system reduce errors in your workflow?	Closed-ended	Confirms operational benefits and validates system effectiveness.
System Admin	What data should the system store to support travel planning?	Open-ended	Defines core data requirements that directly influence database structure and entity design.
System Admin	Should the system process visa applications directly?	Closed-ended	Clarifies system boundaries and ensures proper scope definition.
System Admin	What are the main limitations the system should respect?	Open-ended	Identifies constraints that will shape system design and scope.
System Admin	Is scalability a concern for this system in the future?	Closed-ended	Confirms whether future growth should be considered in system design.

iii. Formal Interviews

Interview 1 – Frequent Traveler

Frequent Traveler	
Interviewee: Oleg, frequent traveler	Interviewer: Daniyal Zharov
Location / Medium: Online - Zoom Meetings	Appointment Date: 13/04/2026 Start time: 12:00 End Time: 12:25
Objectives: - Understand how users currently plan trips - Identify key problems and frustrations - Determine desired features for TravelEase	Reminders: - Interviewee is a regular traveler with experience using multiple booking platforms - May have strong opinions about existing tools - Keep focus on planning process, not payment or post-travel experience
Agenda: Introduction Background on Project Overview of Interview and Topics to Be Covered Trip Planning Process Questions Problems and Challenges Questions Desired Features Questions ... Summary of Major Points Questions from Interviewee Closing	Approximate Time: 1 minute 2 minutes 1 minute 5 minutes 5 minutes 5 minutes 2 minutes ... 2 minutes 3 minutes 1 minute
Questions and Responses:	
Q1 (Open): How do you currently plan your trips?	Answer: I usually search for flights first on one website, then I check hotels on another, and after that I look up visa requirements separately on a government website. Observation: User relies on at least three separate platforms for a single trip, confirming the fragmentation problem.
Q2 (Open): What problems do you face during this process?	Answer: Everything is separated so I have to remember all the details myself. It gets confusing and takes a lot of time. Observation: Strong emphasis on inefficiency and cognitive load placed on the user.

Q3 (Open): How much time do you typically spend planning a single trip?	Answer: It usually takes me a few hours, sometimes more if I need a visa. Observation: Significant time investment confirms the need for a more efficient solution.
Q4 (Closed): Do you ever miss important information such as visa requirements?	Answer: Yes, it has happened before. I once arrived at the airport and found out I needed a visa I did not apply for. Observation: Real-world consequence of the current system gap. Strongly supports the need for integrated visa information.
Q5 (Closed): Would you prefer a system that combines all travel services in one place?	Answer: Yes, definitely. It would make things much easier. Observation: Clear and direct validation of the core concept behind TravelEase.
General Observations: The interviewee demonstrated strong frustration with the current fragmented planning process. The missed visa incident was a significant finding that directly supports one of TravelEase's key features. Overall, the interview confirmed that integration and centralization are the most valued improvements from a user perspective. Unresolved Issues: Payment handling and post-booking management were not discussed and may require follow-up in a later interview or survey.	

Interview 2 – Travel Agency Booking Staff

Travel Agency Booking Staff	
Interviewee: Travel Agency Staff Member	Interviewer: Daniyal Zharov
Location / Medium: Online – Zoom Meetings	Appointment Date: 13/04/2026 Start time: 12:35 End Time: 13:00
Objectives: • Understand how bookings are managed from a service perspective • Identify common issues in managing travel data • Determine what information is essential for a booking system	Reminders: • Interviewee works with booking systems daily • May be familiar with data management challenges • Focus on data and workflow, not customer service topics

Agenda: Introduction Background on Project Overview of Interview and Topics to Be Covered Booking Workflow Questions Common Issues Questions System Value Questions ... Summary of Major Points Questions from Interviewee Closing	Approximate Time: 1 minute 2 minutes 1 minute 5 minutes 5 minutes 5 minutes 2 minutes ... 2 minutes 3 minutes 1 minute
Questions and Responses:	
Q1 (Open): What information is required when organizing a trip for a customer?	Answer: We need customer details, flight information, hotel bookings, and sometimes visa information depending on the destination. Observation: Directly confirms the core data entities the system needs to store.
Q2 (Open): What are the most common issues you encounter during the booking process?	Answer: Customers often change their plans, and managing everything across different systems is difficult and time-consuming. Observation: Highlights the need for a flexible, centralized data management approach.
Q3 (Open): How do you currently manage customer bookings across different platforms?	Answer: We use spreadsheets and different websites. There is no single system that connects everything. Observation: Confirms the absence of integration even at the professional service level.
Q4 (Closed): Would a centralized system reduce errors in your workflow?	Answer: Yes, it would definitely reduce mistakes and save a lot of time. Observation: Validates the operational value of TravelEase from the service side.
General Observations: The booking staff member confirmed that the lack of integration is a problem not only for individual travelers but also at the professional service level. The reliance on spreadsheets and disconnected platforms is a strong indicator that a centralized system would provide real operational value. Unresolved Issues: Automation of booking updates and notification systems were not discussed and could be explored in future requirement-gathering sessions.	

Interview 3 – System Administrator

System Administrator	
Interviewee: System Administrator	Interviewer: Daniyal Zharov
Location / Medium: Online – Zoom Meetings	Appointment Date: 13/04/2026 Start time: 17:00 End Time: 17:30
Objectives: - Identify core data requirements and system structure - Define system boundaries and limitations - Confirm scalability and future considerations	Reminders: - Interviewee has a technical background in system design - May raise concerns about scope and feasibility - Focus on data, structure, and constraints
Agenda: Introduction Background on Project Overview of Interview and Topics to Be Covered Data Requirements Questions System Boundaries Questions Scalability and Constraints Questions ... Summary of Major Points Questions from Interviewee Closing	Approximate Time: 1 minute 2 minutes 1 minute 5 minutes 5 minutes 5 minutes 2 minutes ... 2 minutes 3 minutes 1 minute
Questions and Responses:	
Q1 (Open): What data should the system store to support travel planning?	Answer: The system should store user profiles, trip records, flight and hotel bookings, destination information, and visa-related data. Observation: Provides a clear and complete list of core data entities for the system
Q2 (Closed): Should the system process visa applications directly?	Answer: No, it should only provide guidance and information. Processing applications involves legal and governmental processes that are outside the system's scope. Observation: Clearly defines an important system boundary confirmed from the technical perspective.

Q3 (Open): What are the main limitations the system should respect?	Answer: It should not interact with external government databases directly, and it should not handle financial transactions. Observation: Further refines system boundaries and reduces development complexity.
Q4 (Closed): Would a centralized system reduce errors in your workflow?	Answer: Yes, ideally the system should be built in a way that allows new features or destinations to be added later. Observation: Suggests the database and system design should be flexible and extensible from the start.
General Observations: The system administrator provided the most technically grounded perspective of the three interviews. The responses confirmed the core data entities identified in earlier interviews and added important boundary definitions. The scalability point introduces a design consideration that should be carried forward into the database design phase. Unresolved Issues: Authentication, user roles, and access control were not discussed and should be addressed during the system design phase.	

b. Requirement Structuring

i. Process Modelling - Data Flow Diagrams

Context DFD

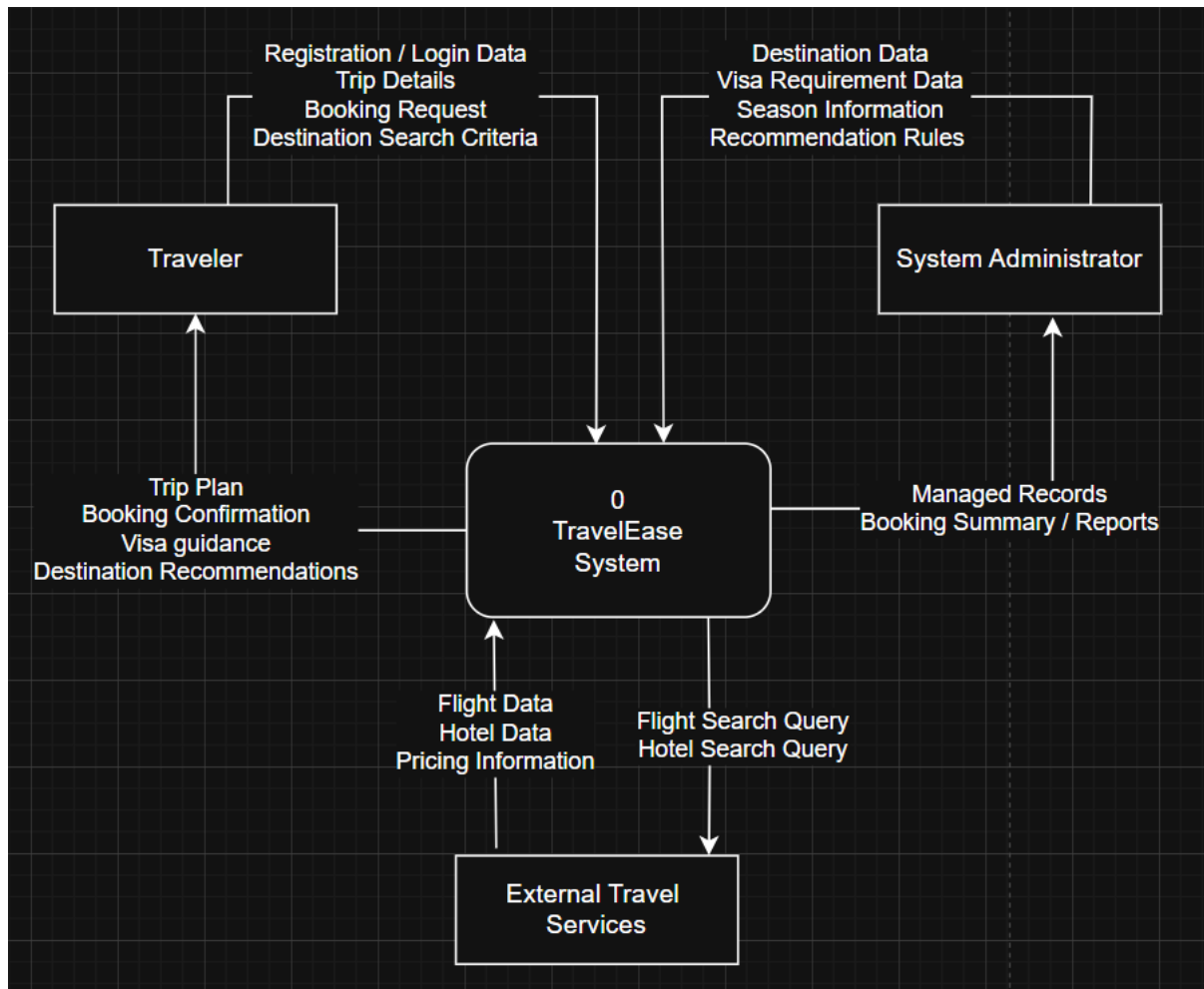
The context DFD represents the entire TravelEase system as a single process that interacts with three other entities: the Traveler, the System Administrator, and External Travel Services. At this level of the DFD, there are no data stores for the system, and only the information flows between the system and the outside environment are represented.

The Traveler is the main user of the system. The system receives data from the Traveler such as their registration/ login information, the details of the trips that they would like to take, and the criteria for the searches for destinations. Based on this information, the system provides the Traveler with plan information for those trips, booking confirmations, information regarding visas for those countries, and information regarding destinations.

The System Administrator manages the content of the system. The System Administrator provides information to the system regarding destinations, visas required for those destinations, and information regarding the best time of year to visit those destinations. The system provides the System Administrator with the records of the bookings made by travelers, as well as any reports regarding those bookings.

External Travel Services represents external providers of information for the system, such as flight and hotel data providers. These external services provide the system with information regarding available flights and available hotels. The system sends requests to these external services based on the information provided by the travelers, and receives a response with the results of those requested searches.

Figure 1. Context-Level Data Flow Diagram (DFD) of the TravelEase System



The diagram adheres to all standard DFD rules at the context level. It contains exactly one process representing the entire system, three external entities that interact with the system from outside its boundary, and only major data flows with no internal processes or data stores shown. The inclusion of External Travel Services as a third entity ensures consistency with the system description in Phase 1, where TravelEase was defined as a system that aggregates data from external providers rather than owning it directly.

Level-0 DFD

The Level-0 DFD for the TravelEase system reveals four internal processes that manage the user's travel experience. Unlike the Context DFD, this diagram illustrates the internal processes that handle the data for the system.

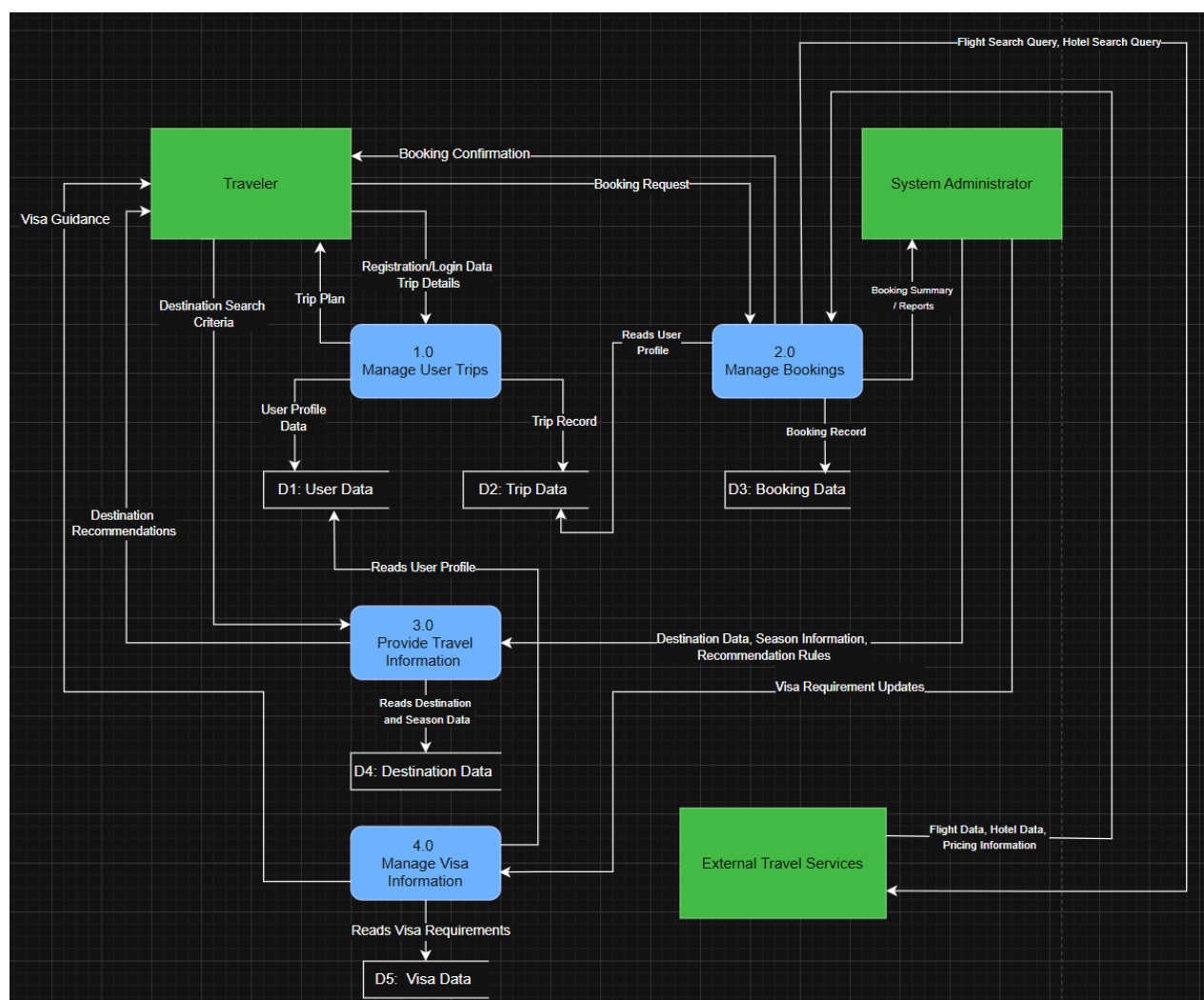
Process 1.0, Manage User Trips, accepts login and trip details from the Traveler and stores them in the system. This process stores the information in D1 User Data and D2 Trip Data. Additionally, it outputs a trip plan for the traveler after storing their data.

Process 2.0, Manage Bookings, accepts booking requests from the Traveler. This process reads the traveler's information from D1 User Data. It also stores the booking request in D3 Booking Data and forwards flight and hotel search requests to the External Travel Services. Finally, it outputs a confirmation message to the Traveler and sends booking summaries to the System Administrator.

Process 3.0, Provide Travel Information, receives travel search requests from the Traveler. This process reads information from D4 Destination Data programmed by the System Administrator and displays the results to the Traveler.

Process 4.0, Manage Visa Information, retrieves visa requirements according to the destination of the Traveler's choice. This process reads from D5 Visa Data programmed by the System Administrator and displays the results to the Traveler.

Figure 2. Level-0 Data Flow Diagram (DFD) of the TravelEase System



The diagram follows all standard DFD rules at the Level-0 level. The four processes cover all major functional areas of the system identified during the requirement determination phase. The five data stores reflect the core data entities that will later be formalized in the Entity-Relationship Diagram. All external entities from the Context DFD are carried forward without modification, ensuring consistency across DFD levels. Solid arrows represent write or send operations, while dashed arrows represent read operations from data stores.

Level-1 DFD — Process 2.0 Manage Bookings

The Level-1 DFD decomposes Process 2.0 Manage Bookings from the Level-0 DFD into four sub-processes. This is the primitive level of decomposition for TravelEase.

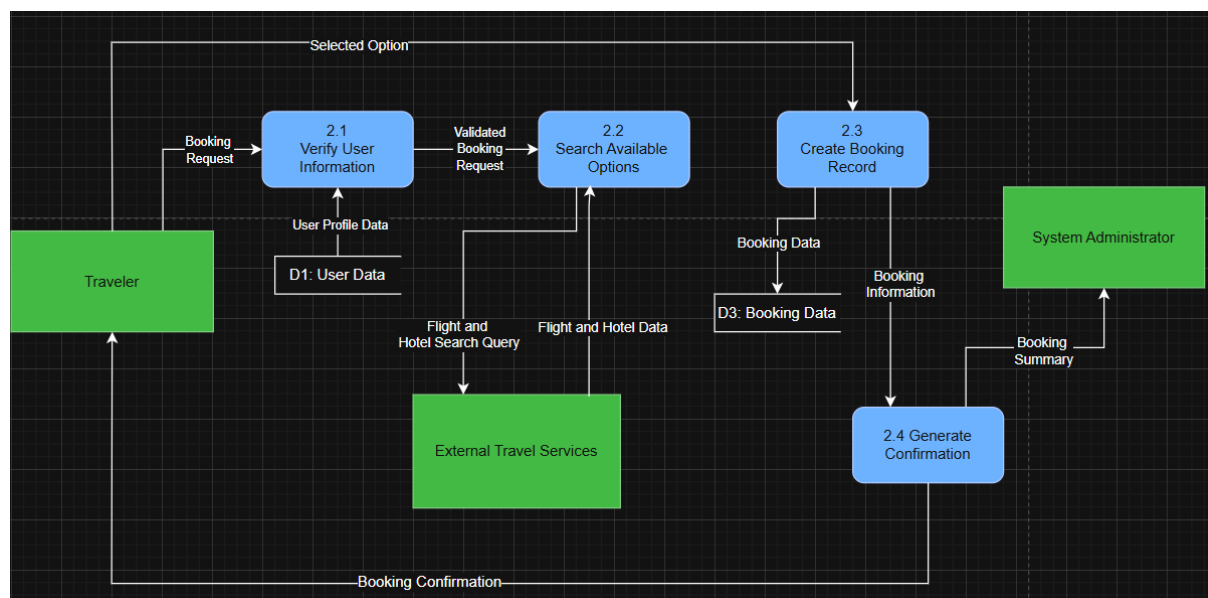
Process 2.1 Verify User Information receives the booking request from the Traveler and reads the user's profile from D1 User Data. If the user is found in the database, the booking request is forwarded to the next process with the status of "validated."

Process 2.2 Search Available Options takes the validated booking request and sends search queries for flights and hotels to the External Travel Services. The External Travel Services send back available flights, available hotels, and the prices for each. This information is sent to the next process.

Process 2.3 Create Booking Record takes the available flights, available hotels, and prices from Process 2.2 and receives the Traveler's choice of flights and hotels. Based on this information, a booking is created and written to D3 Booking Data. The booking information is sent to the next process.

Process 2.4 Generate Confirmation takes the booking information from Process 2.3 and creates a booking confirmation for the Traveler. Additionally, a summary of the booking is sent to the System Administrator.

Figure 3. Level-1 Data Flow Diagram (DFD) — Explosion of Process 2.0 Manage Bookings



The Level-1 DFD follows all standard DFD rules. The external entities and data stores are carried forward from the Level-0 DFD without modification, ensuring consistency across levels. Only the data stores and external entities that Process 2.0 interacts with are included at this level, namely D1

User Data, D3 Booking Data, External Travel Services, the Traveler, and the System Administrator. The decomposition is balanced, with each sub-process handling one clearly defined responsibility within the booking workflow.

iii. Logic Modeling: Decision Tables

Decision tables can be used to represent the logic of processes that involve a number of conditions and actions associated with those conditions. Two processes that involve logic that can be represented within a decision table are Process 2.3 Process Booking and Process 4.0 Manage Visa Information.

Decision Table 1 – Process 2.3: Create Booking Record

This decision table represents the decision logic used in the booking process. Three conditions are evaluated: whether the user is verified, whether booking options are available, and whether the user has selected an option. A booking record is created only when all three conditions are satisfied. Otherwise, the system responds with the appropriate action depending on which condition is not met.

Conditions

Conditions	R1	R2	R3	R4
User verified?	Y	Y	Y	N
Options available?	Y	Y	N	-
Option selected?	Y	N	-	-

Actions

Actions	R1	R2	R3	R4
Create booking record	✓	✗	✗	✗
Request user selection	✗	✓	✗	✗
Display no options message	✗	✗	✓	✗
Redirect to login	✗	✗	✗	✓

Simplified Decision Table 1

Conditions	R1	R2
All conditions satisfied?	Y	N

Actions	R1	R2
Create booking record	✓	✗
Display appropriate message	✗	✓

The simplified table consolidates rules R2, R3, and R4 into a single negative case, since all of them result in the booking not being created. The detailed distinctions between different failure scenarios are preserved in the full table above.

Decision Table 2 – Process 4.0: Manage Visa Information

This decision table represents the logic used when retrieving visa information for a traveler. Three conditions are evaluated: whether visa data exists for the selected destination, whether the traveler's nationality requires a visa, and whether the visa data is valid and up-to-date. Based on these conditions, the system determines the appropriate response to present to the user.

Conditions

Conditions	R1	R2	R3	R4
Visa data exists for destination?	Y	Y	Y	N
Traveler requires visa?	Y	Y	N	-
Visa data valid and up-to-date?	Y	N	-	-

Actions

Actions	R1	R2	R3	R4
Display visa requirements	✓	✗	✗	✗
Display outdated data warning	✗	✓	✗	✗
Display visa-free confirmation	✗	✗	✓	✗
Display no data available message	✗	✗	✗	✓

Simplified Decision Table 2

Conditions	R1	R2
Valid visa data available and applicable?	Y	N

Actions	R1	R2
Display visa information	✓	✗
Display warning or unavailable message	✗	✓

The simplified table combines rules R2, R3, and R4 into a single negative case, as all of these scenarios prevent the system from providing complete and reliable visa guidance. The detailed table above preserves the specific system response for each condition.

c. Conceptual modelling

The conceptual ERD identifies the core entities of the TravelEase system, their attributes, and the relationships between them. The entities were derived directly from the data stores identified in the Level-0 DFD and the requirements gathered during the analysis phase.

Entities and Attributes

- User (UserID, Name, Email, Password, Nationality)
- Trip (TripID, TripName, StartDate, EndDate, Status)
- Booking (BookingID, BookingType, BookingDate, Status, TotalPrice)
- Flight (FlightID, Airline, DepartureCity, ArrivalCity, DepartureTime, ArrivalTime, Price)
- Hotel (HotelID, HotelName, Location, CheckInDate, CheckOutDate, PricePerNight)
- Destination (DestinationID, CountryName, City, Season, Description)
- Visa (VisaID, VisaType, Requirements, ProcessingTime, EligibleNationalities)

Relationships and Cardinalities

- User creates Trip (1..1 → 0..N)
- Trip contains Booking (1..1 → 0..N)
- Booking includes Flight (0..1 → 1..N)
- Booking includes Hotel (0..1 → 1..N)
- Trip has Destination (0..N → 1..1)
- Destination has Visa (1..1 → 0..N)

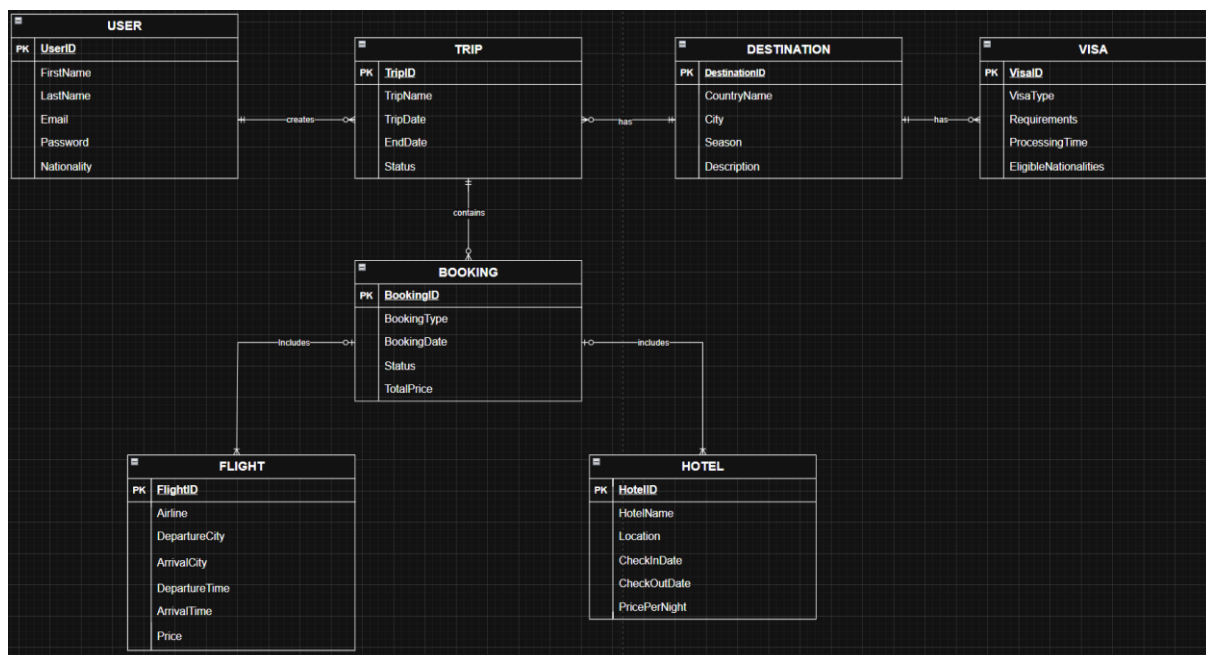


Figure 4. Conceptual Entity-Relationship Diagram – TravelEase System

ERD Description

The User entity is the starting point of all system interactions, as users create and manage trips. Each trip connects to a single destination and groups one or more bookings. The BookingType attribute determines whether a booking is associated with a flight or a hotel, and each booking represents a single service type. The Destination entity links to Visa, ensuring country-specific requirements are accessible based on traveler nationality. The cardinalities are defined as follows: User to Trip (1..1 to 0..N), Trip to Booking (1..1 to 0..N), Booking to Flight (0..1 to 1..N), Booking to Hotel (0..1 to 1..N), Trip to Destination (0..N to 1..1), and Destination to Visa (1..1 to 0..N).

Phase 3 – System Design

a. Designing the Human Interface

i. Designing Forms and Reports

Narrative Overview

The forms and reports in the system are designed to allow travelers to efficiently plan and manage their trips. The primary target audience for the system is individual travelers looking to plan their trips. Since the system is web-based, the interfaces were to be as simple and easy to navigate as possible.

There are three main interface components to the system: a trip creation form, a booking form, and a booking confirmation report. The trip creation form allows users to enter their trip information. The booking form allows users to select either a flight or hotel booking for their selected trip. The booking confirmation report displays the details for each of the user's selected trips and bookings, including visa information and the total cost of the selected trips.

Each of these interfaces follows the same basic principles of good interface design as described in Table 8-2. Each component of the interfaces has a title, includes only the information that is necessary for the user to understand their trips and bookings, and is laid out in a clear and easy to navigate fashion.

Sample Design

Form 1 — Trip Creation Form

The Trip Creation Form consists of a series of labeled input fields for information such as Trip Name, Destination, Start Date, End Date, and Status. These fields are arranged in a vertical list for the user. The User ID field is auto-filled with the logged in user's ID. A visa information panel is displayed automatically after the user selects a destination. Save Trip, Back, and Cancel buttons are displayed at the bottom of the form.

Figure 5. Trip Creation Form — TravelEase System

TravelEase Version 1.0

Trip Creation Form

Create a new trip and select your destination

Page: 1 of 1
Date: 17-APR-2026

User ID: Auto-filled from session

Trip Name: * e.g. Summer Vacation 2026

Destination: * Select destination...

Travel Dates: * Start date End date

Status: Planning / Confirmed / Completed

* Required fields

Visa Information
Based on your selected destination, the system will automatically display applicable visa requirements below.

Visa details will appear here after destination is selected

Cancel Back Save Trip

Form 2 - Booking Form

The Booking Form allows users to select a Booking Type - either Flight or Hotel. The Trip ID and Destination are auto-filled from the previously created trip, ensuring consistency and reducing data entry effort. Based on the booking type selection, the form displays only the relevant fields. For a flight booking, fields such as Departure City, Arrival City, Travel Dates, and Number of Passengers are shown. For a hotel booking, the corresponding hotel fields are displayed instead. This conditional

layout reduces confusion and ensures that users only see necessary inputs. A Search button at the bottom submits the query to External Travel Services and advances the user to the results page.

Figure 6. Booking Form — TravelEase System

The screenshot displays the 'TravelEase' Booking Form, Version 1.0. The form is titled 'Booking Form' with the subtitle 'Search and book flights or hotels for your trip'. It is on 'Page: 1 of 2' with a 'Date: 17-APR-2026'. The form is divided into several sections: 'TRIP REFERENCE' with fields for 'Trip ID' and 'Destination', both auto-filled from a selected trip; 'BOOKING TYPE' with radio buttons for 'Flight' (selected) and 'Hotel'; 'FLIGHT SEARCH CRITERIA' with fields for 'Departure City' (e.g. Dubai), 'Arrival City' (e.g. London), 'Travel Dates' (split into 'Departure date' and 'Return date'), and 'Passengers' (1). A note states '* Required fields | Hotel search fields will appear when Hotel is selected above'. At the bottom are 'Cancel', 'Back', and 'Search →' buttons.

TravelEase		Version 1.0
Booking Form		Page: 1 of 2 Date: 17-APR-2026
Search and book flights or hotels for your trip		
TRIP REFERENCE		
Trip ID:	Auto-filled from selected trip	
Destination:	Auto-filled from selected trip	
BOOKING TYPE		
☒ Flight ☐ Hotel		
FLIGHT SEARCH CRITERIA		
Departure City: *	e.g. Dubai	
Arrival City: *	e.g. London	
Travel Dates: *	Departure date	Return date
Passengers: *	1	
* Required fields Hotel search fields will appear when Hotel is selected above		
Cancel	Back	Search →

Report - Booking Confirmation Report

The Booking Confirmation Report is a system-generated output displayed after a booking is successfully processed. It presents trip information, traveler details, full booking details, booking status, total price, and applicable visa information in a structured table format. Alternating row shading improves readability. The report includes a generated date and a valid-as-of date to satisfy the

meaningful title requirements of Table 8-2. Navigation buttons at the bottom allow the user to return to the dashboard or print the report.

Figure 7. Booking Confirmation Report — TravelEase System

TravelEase

Version 1.0

Booking Confirmation Report

Summary of confirmed booking details

Page: 1 of 1
Generated: 17-APR-2026
Valid as of: 17-APR-2026

TRAVELER INFORMATION

Traveler Name:

John Doe

User ID:

USR-00142

Nationality:

American

TRIP DETAILS

Trip Name:

Summer Vacation 2026

Destination:

London, United Kingdom

Start Date:

01-JUN-2026

End Date:

15-JUN-2026

BOOKING DETAILS

Field	Details
Booking ID:	BKG-20260417-001
Booking Type:	Flight
Airline:	Emirates
Departure:	Dubai (DXB) — 01-JUN-2026 08:00
Arrival:	London Heathrow (LHR) — 01-JUN-2026 12:30
Booking Date:	17-APR-2026
Status:	Confirmed
Total Price:	USD 850.00

VISA INFORMATION

Based on your nationality (American), a valid passport is required for entry into the United Kingdom. No visa is required for stays under 6 months.

Back to Dashboard

Print

Done

Usability Testing

The forms and report were evaluated based on usability factors including clarity, ease of use, and navigation flow. The goal was to ensure that users can complete tasks quickly without confusion or errors.

The Trip Creation Form is easy to use because the fields are arranged in a logical order and require minimal input. The Booking Form improves usability by showing only relevant fields depending on the booking type selected, reducing unnecessary complexity. The Booking Confirmation Report is easy to read because information is clearly grouped into sections and presented in a structured format.

To further improve usability, the system includes validation for required fields and provides clear feedback when incorrect or incomplete data is entered. Date fields include format hints to prevent input errors. Overall, the interface design supports efficient user interaction and aligns with the goal of reducing planning errors and improving the user experience.

ii. Designing Dialogues

Dialogue Diagram

The dialogue diagram shows the dialogue between the user and the TravelEase system, as well as the different screens that the user navigates between to accomplish different tasks. Each of the displays in the system is represented in the dialogue diagram as a three-section box containing the reference number for that display, the name of that display, and the reference numbers of the displays from which the user can return to view the tasks.

The dialogue for the system begins at Display 0, the login screen for the system. Upon successfully logging into the system, the user is directed to Display 1, the dashboard for the system. From the dashboard, the user can navigate to either the Trip Creation Form (Display 2), the Booking Form (Display 3), or the View Trips screen (Display 4).

From Display 2, the user is directed back to Display 2.1 after completing their trip form and content entries. From here, the user can return to either the Dashboard or to the Trip Creation Form screen. From Display 3, the user is directed to Display 3.1 after submitting their travel search. From Display 3.1, the user is directed to Display 3.1.1 which contains the booking processing screen, followed by Display 3.1.1.1 which is the Booking Confirmation screen for the selected trips. From either of these screens, the user can navigate to any screen within the steps taken to reach that screen (to the dashboard, or to the previous screen), as well as back to the Booking Form screen. Finally, at Display 4, the user can browse for available trips to add to their travel plans, or click on the details for any of those available trips to access Display 4.1 which displays more details regarding that potential trip.

Dialogue Design Guidelines

The dialogue diagram for the TravelEase website follows the guidelines defined in Table 8-14. The system is designed to ensure that the user navigates through each step in order, and that each screen within the system has a single defined purpose. Additionally, each screen displays the reference numbers of the screens from which the user can return to, indicating that the user is always able to navigate backwards through their navigations within the system. Finally, each screen is designed with the same general structure and navigation; the screens are provided with feedback after each step (especially upon booking trips), and the steps are defined and clear for the users.

Figure 8. Dialogue Diagram — TravelEase System



iii. Anticipating and Handling Input Errors

The following techniques for handling input errors are applied to the TravelEase system, as summarized in Tables 8-7 through 8-11.

First, the system follows the guidelines for data-entry design (Table 8-7). The system avoids asking for information that can be automatically filled in, such as the User ID and Trip ID fields. Other fields have default values for their entries, such as pre-selected options for booking types. All fields are appropriately labeled with descriptions of the type of information that is required for each field.

Second, the system incorporates techniques for effective data input formatting (Table 8-8). The fields include clear captions for the fields next to the fields themselves, and some fields have specific formats applied to them. For instance, fields for dates have specific date formats requested for those fields of entries, and fields that have limited options for the choices that can be selected have radio buttons for those options.

Third, the system avoids common data entry errors (Table 8-9). The controls for the system are designed to avoid common entry errors for fields, such as preventing the user from entering characters into a field that only accepts numbers, or forcing entries into fields that only have predefined entries for those fields.

Fourth, the system uses validation techniques before it saves the data (Table 8-10). Some of these techniques include making sure that there is no missing data for required fields, ensuring that entries for fields like dates or the number of passengers falls within a certain range, ensuring that data entered has the appropriate format for those fields, and making sure that the data entered into fields is still reasonable in the context of the form, such as preventing the user from choosing the same cities for departure and arrival.

Finally, the system displays user-friendly error messages (Table 8-11). Any errors that occur during data entry are described in a way that a user can understand, as well as with instructions for how to correct that error. For instance, if an error indicates that a user has incorrectly entered a date, the error message may indicate the specific type of invalid entry by the user, while if a required field is missing data, the system will highlight that field and indicate that data is required for that field.

Overall, the implementation of these different techniques will ensure that the system is efficient to use, accurate in the data that it enters from the users, and that the likelihood of errors from those users is minimized.

b. Logical Modelling

i. Relational Model Preparation

The relational model was developed by following a structured approach based on the conceptual ERD produced in Phase 2. The model was expanded during the logical modeling phase to include additional entities that support the full functionality of the TravelEase system. These additions include the Country, Season, VisaRequirement, and Recommendation entities, which provide a more complete and realistic representation of the system's data requirements.

ii. Detailed ERD

Entities and Attributes

The following entities and their attributes were identified based on the conceptual model and expanded during logical modeling:

- USER (UserID, FirstName, LastName, Email, PasswordHash, Nationality)
- TRIP (TripID, TripName, StartDate, EndDate, Status, UserID, DestinationID)
- BOOKING (BookingID, BookingType, BookingDate, Status, TotalPrice, TripID)
- FLIGHT (FlightID, Airline, DepartureCity, ArrivalCity, DepartureTime, ArrivalTime, Price, BookingID)
- HOTEL (HotelID, HotelName, Location, CheckInDate, CheckOutDate, PricePerNight, BookingID)
- DESTINATION (DestinationID, Description, City, CountryCode)
- COUNTRY (CountryCode, CountryName, Region, Currency)
- SEASON (SeasonID, SeasonName, DestinationID)
- VISAREQUIREMENT (CountryCode, VisaType, Requirements, ProcessingTime)
- RECOMMENDATION (DestinationID, SeasonID, RecCategory, RecDescription, PopularityScore)

Each entity contains a primary key, and foreign keys are used to enforce relationships between entities.

Attribute Types

Most attributes across all entities are simple single-valued attributes. The exceptions are noted below:

- **Primary Keys:** UserID, TripID, BookingID, FlightID, HotelID, DestinationID, CountryCode, SeasonID
- **Composite Primary Keys:** VISAREQUIREMENT (CountryCode + VisaType), RECOMMENDATION (DestinationID + SeasonID)
- **Foreign Keys:** UserID and DestinationID in TRIP, TripID in BOOKING, BookingID in FLIGHT and HOTEL, CountryCode in DESTINATION, DestinationID in SEASON, CountryCode in VISAREQUIREMENT, DestinationID and SeasonID in RECOMMENDATION
- **Derived:** TotalPrice in BOOKING is calculated based on the associated flight or hotel price

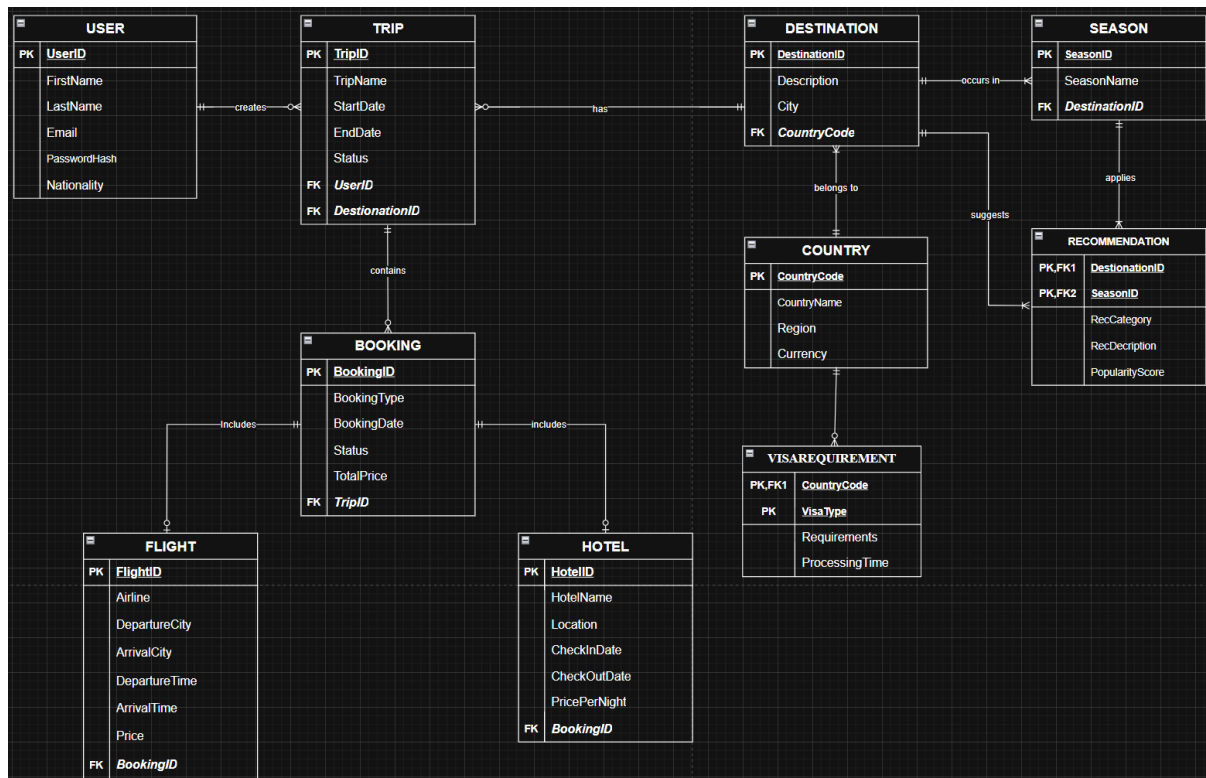


Figure 9. Detailed Entity-Relationship Diagram – TravelEase System

Relationships and Cardinalities

User creates Trip: (1..1) to (0..N): a user can create many trips, each trip belongs to exactly one user

Trip contains Booking: (1..1) to (0..N): a trip can have many bookings, each booking belongs to exactly one trip

Booking includes Flight: (0..1) to (0..1): a booking may include one flight, a flight belongs to at most one booking

Booking includes Hotel: (0..1) to (0..1): a booking may include one hotel, a hotel belongs to at most one booking

Trip has Destination: (0..N) to (1..1): many trips can share the same destination, each trip has exactly one destination

Destination belongs to Country: (0..N) to (1..1): many destinations can belong to the same country, each destination belongs to exactly one country

Country defines VisaRequirement: (1..1) to (0..N): a country can have many visa requirements, each visa requirement belongs to exactly one country

Destination occurs in Season: (1..1) to (0..N): a destination can have many seasonal records, each season record belongs to exactly one destination

Destination suggests Recommendation: (1..1) to (0..N): a destination can have many recommendations, each recommendation belongs to exactly one destination

Season applies to Recommendation: (1..1) to (0..N): a season can apply to many recommendations, each recommendation is linked to exactly one season

Domain Values

Attribute	Domain Values
Status (Trip)	Planning, Confirmed, Completed
BookingType	Flight, Hotel
SeasonName	Winter, Spring, Summer, Fall
PopularityScore	Integer between 1 and 100
Email	Valid email format (e.g., user@domain.com)
CountryCode	ISO 3166-1 alpha-2 format (e.g., AE, GB, US)
StartDate / EndDate	Valid calendar date, StartDate must be before EndDate
TotalPrice	Decimal value greater than 0
DepartureTime / ArrivalTime	Valid datetime, DepartureTime must be before ArrivalTime
PricePerNight / Price	Decimal value greater than 0
CheckInDate / CheckOutDate	Valid calendar date, CheckInDate must be before CheckOutDate

iii. Shorthand Relational Model

Primary keys are underlined and foreign keys are italicized in the report. The shorthand notation is as follows:

- USER (UserID, FirstName, LastName, Email, PasswordHash, Nationality)
- TRIP (TripID, TripName, StartDate, EndDate, Status, *UserID*, *DestinationID*)
- BOOKING (BookingID, BookingType, BookingDate, Status, TotalPrice, *TripID*)
- FLIGHT (FlightID, Airline, DepartureCity, ArrivalCity, DepartureTime, ArrivalTime, Price, *BookingID*)
- HOTEL (HotelID, HotelName, Location, CheckInDate, CheckOutDate, PricePerNight, *BookingID*)
- DESTINATION (DestinationID, Description, City, *CountryCode*)
- COUNTRY (CountryCode, CountryName, Region, Currency)
- SEASON (SeasonID, SeasonName, *DestinationID*)
- VISAREQUIREMENT (CountryCode, VisaType, Requirements, ProcessingTime)
- RECOMMENDATION (*DestinationID*, *SeasonID*, RecCategory, RecDescription, PopularityScore)

iv. Normalization to 3NF

All relations satisfy Third Normal Form. Each table satisfies First Normal Form as all attributes are atomic and there are no repeating groups. Each table satisfies Second Normal Form as all non-key attributes are fully functionally dependent on the entire primary key with no partial dependencies present. Third Normal Form is satisfied as there are no transitive dependencies — no non-key attribute depends on another non-key attribute. The schema is therefore fully normalized to 3NF with no further modification required.

v. Table Merging

The tables derived from the interface designs in Step C(a) are consistent with the tables produced in Step C(b)iii. The Trip Creation Form maps to the TRIP and DESTINATION tables, the Booking Form maps to the BOOKING, FLIGHT, and HOTEL tables, and the Booking Confirmation Report draws from the BOOKING, TRIP, USER, and VISAREQUIREMENT tables. No structural conflicts were identified between the interface-derived tables and the ERD-derived tables, confirming that both sources are fully aligned.

vi. Merging Problems

Table merging was considered but not performed for the following reasons. Merging FLIGHT and HOTEL into BOOKING would introduce null values for every record since a booking is always either a flight or a hotel but never both simultaneously, violating the principle of avoiding null-heavy tables and reducing data integrity. Merging COUNTRY with VISAREQUIREMENT would create redundancy since country information would be duplicated across multiple visa records. Merging RECOMMENDATION with DESTINATION would risk violating 1NF by introducing multivalued attributes, as a destination can have multiple recommendations across different seasons.

If merging were performed, the potential problems would include increased data redundancy, the introduction of null values in non-applicable fields, violations of normalization principles, and reduced query efficiency. The decision not to merge preserves the integrity and scalability of the schema.

vii. Final Normalization

Since no tables were merged, the schema remains fully in 3NF. The structure preserves data integrity, avoids redundancy, and supports efficient querying across all system operations. All primary and foreign key relationships are intact and all domain constraints defined in Section C(b)ii are enforced at the schema level.

c. Physical Data Modeling

i. Database Implementation

The TravelEase database was implemented using **MariaDB** as the chosen Database Management System. MariaDB was selected due to its compatibility with the relational model defined in Phase 3b, its support for ENUM data types for enforcing domain constraints, its robust foreign key enforcement via InnoDB engine, and its widespread use in web-based application development.

All ten normalized tables were created with appropriate data types, default values, domain constraints, and foreign key relationships. The implementation directly reflects the shorthand relational model defined in Section C(b)iii.

DBMS: MariaDB **Database Name:** TravelEase **Storage Engine:** InnoDB **Character Set:** utf8mb4

Data Types and Domain Ranges:

Table	Attribute	Data Type	Constraint
USER	UserID	INT AUTO_INCREMENT	PRIMARY KEY
USER	Name	VARCHAR(100)	NOT NULL
USER	Email	VARCHAR(150)	UNIQUE NOT NULL
USER	Nationality	VARCHAR(50)	NOT NULL
TRIP	Status	ENUM	Planning, Confirmed, Completed
TRIP	StartDate / EndDate	DATE	StartDate must be before EndDate
BOOKING	BookingType	ENUM	Flight, Hotel
BOOKING	TotalPrice	DECIMAL(10,2)	DEFAULT 0
COUNTRY	CountryCode	CHAR(2)	PRIMARY KEY — ISO 3166-1 alpha-2
VISAREQUIREMENT	CountryCode + VisaType	CHAR(2) + VARCHAR(50)	COMPOSITE PRIMARY KEY
RECOMMENDATION	PopularityScore	INT	CHECK BETWEEN 1 AND 100

Figure 10. TravelEase Database Tables — MariaDB Implementation via HeidiSQL

The screenshot shows the 'travelease' database with a total size of 304.0 KiB. It contains ten tables with the following sizes:

Table Name	Size
booking	32.0 KiB
country	16.0 KiB
destination	32.0 KiB
flight	32.0 KiB
hotel	32.0 KiB
recommendation	32.0 KiB
season	32.0 KiB
trip	48.0 KiB
user	32.0 KiB
visarequirement	16.0 KiB

Figure 11. BOOKING Table Structure — Data Types and Constraints in HeidiSQL

The screenshot shows the 'booking' table structure with the following columns and constraints:

#	Name	Datatype	Length/Set	Unsigned	Allow N...	Zerofill	Default	Comment	Collation	Expression	Virtuality
1	BookingID	INT	11	☐	☐	☐	AUTO_INCREME...				
2	BookingType	ENUM	'Flight','Hotel'	☐	☒	☐	NULL		utf8mb4_general_ci		
3	BookingDate	DATE		☐	☒	☐	NULL				
4	Status	VARCHAR	50	☐	☒	☐	NULL		utf8mb4_general_ci		
5	TotalPrice	DECIMAL	10,2	☐	☒	☐	'0.00'				
6	TripID	INT	11	☐	☒	☐	NULL				

Figure 12. VISAREQUIREMENT Table Structure — Composite Primary Key in HeidiSQL

The screenshot shows the 'visarequirement' table structure with the following columns and constraints:

#	Name	Datatype	Length/Set	Unsigned	Allow N...	Zerofill	Default	Comment	Collation	Expression	Virtuality
1	CountryCode	CHAR	2	☐	☐	☐	No default		utf8mb4_general_ci		
2	VisaType	VARCHAR	50	☐	☐	☐	No default		utf8mb4_general_ci		
3	Requirements	TEXT		☐	☒	☐	NULL		utf8mb4_general_ci		
4	ProcessingTime	INT	11	☐	☒	☐	NULL				

The complete SQL script used to create and populate the database is submitted alongside this report as **seed_data.sql**. This file includes all CREATE TABLE statements with data types, constraints, and foreign key definitions, as well as INSERT statements providing seed data for all ten tables.

Phase 4 - Systems Implementation

Linking Interface Designs to Database

To satisfy Phase 4, the interface designs produced in Phase 3a were linked to the TravelEase database through a fully functional web application built using **Python Flask** and **PyMySQL**.

The application consists of five pages that directly correspond to the interface designs and dialogue diagram produced in Phase 3: a Login page, a Dashboard, a Trip Creation Form, a Booking Form, and a Booking Confirmation Report. Each page interacts with the MariaDB database in real time — reading from and writing to the normalized tables defined in Phase 3b.

Technology Stack:

- Backend: Python Flask
- Database Connector: PyMySQL
- Frontend: HTML5 + CSS3
- Database: MariaDB

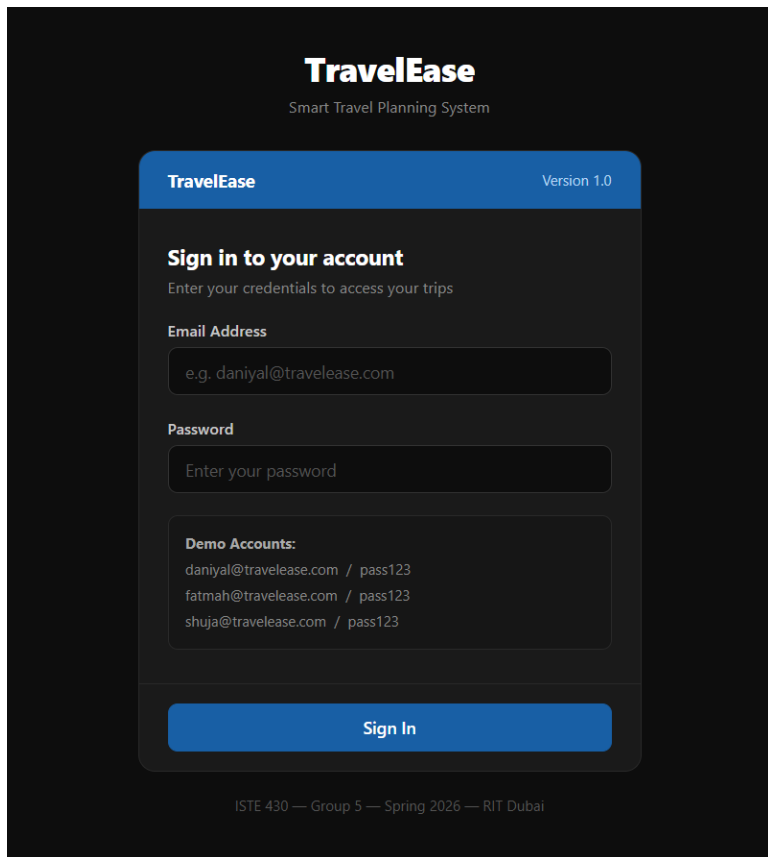
Page-to-Database Mapping:

Page	Route	Database Interaction
Login	/login	Reads from USER table to authenticate
Dashboard	/dashboard	Reads from TRIP, DESTINATION, COUNTRY, BOOKING
Trip Creation Form	/create-trip	Reads DESTINATION, COUNTRY — Inserts into TRIP
Booking Form	/booking	Reads TRIP — Inserts into BOOKING + FLIGHT or HOTEL
Booking Confirmation	/confirmation	Reads BOOKING, TRIP, FLIGHT, HOTEL, USER, VISAREQUIREMENT

Session Management: The application uses Flask session handling to maintain the logged-in user across pages. All database queries are scoped to the authenticated user's UserID, ensuring data isolation between users.

Input Validation: All forms include server-side validation that enforces the domain constraints defined in Phase 3b - required field checks, date range validation, numeric type checks, and logical constraints such as preventing identical departure and arrival cities.

Figure 13. TravelEase Login Page



The login page features a dark blue header with the 'TravelEase' logo and 'Smart Travel Planning System' tagline. Below this is a white login card with a blue header bar containing 'TravelEase' and 'Version 1.0'. The card contains a 'Sign in to your account' section with a subtitle 'Enter your credentials to access your trips'. It includes input fields for 'Email Address' (with a placeholder 'e.g. daniyal@travelease.com') and 'Password' (with a placeholder 'Enter your password'). A 'Demo Accounts' section lists three accounts: 'daniyal@travelease.com / pass123', 'fatmah@travelease.com / pass123', and 'shuja@travelease.com / pass123'. A large blue 'Sign In' button is at the bottom of the card. The footer of the page reads 'ISTE 430 — Group 5 — Spring 2026 — RIT Dubai'.

TravelEase
Smart Travel Planning System

TravelEase Version 1.0

Sign in to your account
Enter your credentials to access your trips

Email Address
e.g. daniyal@travelease.com

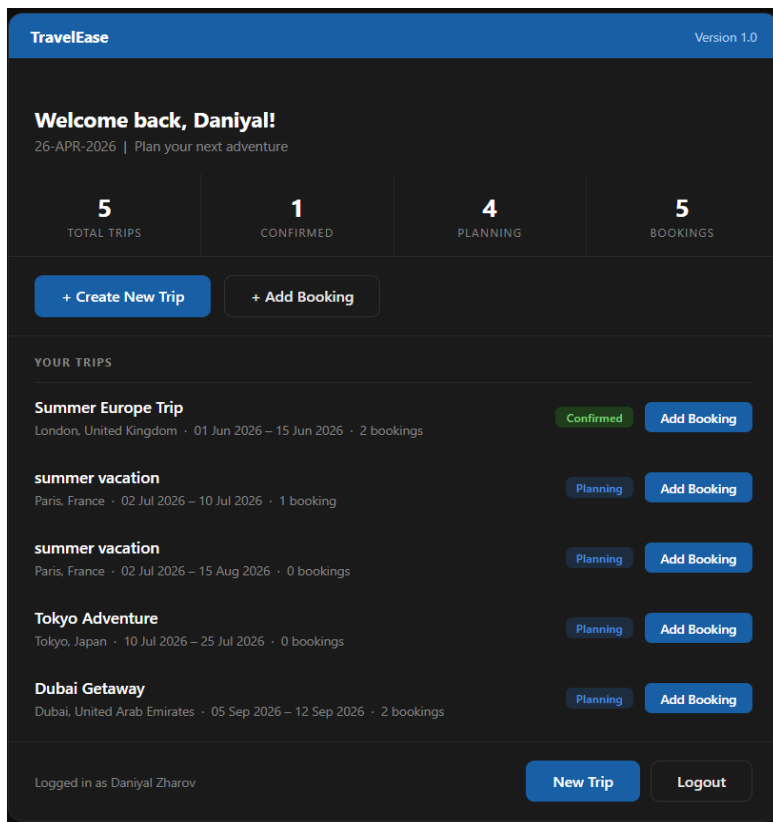
Password
Enter your password

Demo Accounts:
daniyal@travelease.com / pass123
fatmah@travelease.com / pass123
shuja@travelease.com / pass123

Sign In

ISTE 430 — Group 5 — Spring 2026 — RIT Dubai

Figure 14. TravelEase Dashboard



The dashboard has a dark blue header with 'TravelEase' and 'Version 1.0'. Below the header is a 'Welcome back, Daniyal!' message with the date '26-APR-2026' and a link 'Plan your next adventure'. A summary section shows four metrics: '5 TOTAL TRIPS', '1 CONFIRMED', '4 PLANNING', and '5 BOOKINGS'. Below these are two buttons: '+ Create New Trip' and '+ Add Booking'. The 'YOUR TRIPS' section lists five trips with their details and status. The footer shows 'Logged in as Daniyal Zharov' and two buttons: 'New Trip' and 'Logout'.

TravelEase Version 1.0

Welcome back, Daniyal!
26-APR-2026 | Plan your next adventure

5 TOTAL TRIPS
1 CONFIRMED
4 PLANNING
5 BOOKINGS

+ Create New Trip **+ Add Booking**

YOUR TRIPS

Trip Name	Location	Dates	Bookings	Status	Action
Summer Europe Trip	London, United Kingdom	01 Jun 2026 – 15 Jun 2026	2 bookings	Confirmed	Add Booking
summer vacation	Paris, France	02 Jul 2026 – 10 Jul 2026	1 booking	Planning	Add Booking
summer vacation	Paris, France	02 Jul 2026 – 15 Aug 2026	0 bookings	Planning	Add Booking
Tokyo Adventure	Tokyo, Japan	10 Jul 2026 – 25 Jul 2026	0 bookings	Planning	Add Booking
Dubai Getaway	Dubai, United Arab Emirates	05 Sep 2026 – 12 Sep 2026	2 bookings	Planning	Add Booking

Logged in as Daniyal Zharov **New Trip** **Logout**

Figure 15. . TravelEase Trip Creation Form — Connected to Database

TravelEase

Version 1.0

Trip Creation Form

Page: 1 of 1
Date: 26-APR-2026

Create a new trip and select your destination

User ID:

Auto-filled from session

Trip Name: *

e.g. Summer Vacation 2026

Destination: *

Select destination...

Travel Dates: *

dd/mm/yyyy

dd/mm/yyyy

Status:

Planning

* Required fields

Visa Information

Based on your selected destination, the system will automatically display applicable visa requirements below.

Visa details will appear here after destination is selected

Cancel

Back

Save Trip

Figure 16. TravelEase Booking Form — Connected to Database

TravelEase

Version 1.0

Booking Form

Page: 1 of 2
Date: 26-APR-2026

Search and book flights or hotels for your trip

TRIP REFERENCE

Trip ID:

Summer Europe Trip - London, United Kingdom

Destination:

London, United Kingdom

BOOKING TYPE

☒ Flight

☐ Hotel

FLIGHT SEARCH CRITERIA

Airline: *

e.g. Emirates

Departure City: *

e.g. Dubai

Arrival City: *

e.g. London

Travel Dates: *

dd/mm/yyyy --:-- --

dd/mm/yyyy --:-- --

Price (USD): *

e.g. 850.00

* Required fields | Hotel search fields will appear when Hotel is selected above

Cancel

Back

Search

Figure 16 shows the Booking Form running as a live web interface connected to the TravelEase MariaDB database. The Trip dropdown is dynamically populated by querying the TRIP table filtered by the logged-in user's session ID, ensuring that each user only sees their own trips. Upon selecting a booking type, the form displays either flight or hotel input fields. On submission, the system inserts a new record into the BOOKING table and a corresponding record into either the FLIGHT or HOTEL table depending on the selected booking type, with the BookingID used as a foreign key to link the two records. All input is validated server-side before any database write is performed.

Figure 17. TravelEase Booking Confirmation Report — Dynamic Data from Database

TravelEase

Version 1.0

Booking Confirmation Report

Summary of confirmed booking details

Page: 1 of 1

Generated: 26-APR-2026

Valid as of: 26-APR-2026

TRAVELER INFORMATION

Traveler Name:

Daniyal Zharov

User ID:

USR-00001

Nationality:

Kazakhstani

TRIP DETAILS

Trip Name:

Daniyal's Summer Vacation

Destination:

Bangkok, Thailand

Start Date:

13-JUL-2026

End Date:

20-JUL-2026

BOOKING DETAILS

Booking ID:

BKG-20260426-007

Booking Type:

Flight

Airline:

Air Astana

Departure:

Astana - 13-JUL-2026 09:30

Arrival:

Bangkok - 20-JUL-2026 19:55

Booking Date:

26-APR-2026

Status:

Confirmed

Total Price:

USD 400.00

VISA INFORMATION

Tourist Visa

Many nationalities receive 30-day visa-on-arrival.

Back to Dashboard

Print

Done

The complete source code including app.py, all HTML templates, the CSS stylesheet, and the SQL script are submitted alongside this report as required by the project guidelines.

Conclusion

Developing TravelEase through all four phases of the System Development Life Cycle provided an overall understanding of the various stages of system analysis, design, and implementation. Creating a system that fulfills the requirements of a user requires more than just the development of the system's code; requirements and plans must be made for the system prior to its development.

The early stages of creating TravelEase, such as determining the requirements of the system and modeling the data flows within the system, helped to inform the designs of both the system and its database. Thus, these early stages helped to ensure the consistency of the system throughout its creation, from its design to implementation stage.

One of the most important aspects of the system creation process was ensuring the consistency of each component of the system. Whether reviewing the entities for the system, the processes within the system, the interface designs, or the database itself, each of these components must be consistent with one another. Ensuring the consistency of the system created both challenges and an essential component in the creation of the system.

Finally, the implementation of TravelEase as a working Flask web application allowed for each of the features for the system (user authentication, trip creation, flight and hotel booking, visa information retrieval, and more) to be developed and integrated into the system. The integration of the front-end web application interface with the back-end relational database indicates the success of the System Development Life Cycle as a process for the creation of an Information System.

Overall, creating TravelEase has allowed each member of the team to gain experience in the development of the system from start to finish. Additionally, the successful development of the system has increased the confidence of each member in their abilities to create systems for other potential applications.

Future improvements could include enhancing security through proper password hashing, expanding the booking system to support multiple services per trip, and improving the user interface with more dynamic feedback and personalization.

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AI Usage Disclosure

In accordance with the course policy on Generative AI, the team acknowledges the use of Claude AI (Anthropic, Version claude-sonnet-4-6, accessed April 2026) during the development of this report.

How AI was used: The AI tool was used as an assistant throughout the project for the following purposes: structuring written sections, suggesting formatting for DFDs, ERDs, and decision tables, generating initial drafts of interview questions, providing guidance on normalization and SQL syntax, and assisting with the Flask web application implementation in Phase 4.

How output was verified: All AI-generated content was reviewed, edited, and validated by the team against course materials, specifically the Valacich textbook, before inclusion in the report. Diagrams were drawn independently by the team using diagrams.net. SQL code was tested and executed against a live MariaDB database to verify correctness.

How output was integrated: AI output was never copied directly without review. All written sections were rewritten or refined by team members to reflect the team's own understanding and the specific context of the TravelEase system. The AI served as a starting point and sounding board rather than a final source.

Limitations acknowledged: The team recognizes that AI tools can produce inaccurate or inconsistent output. Several suggestions made by the AI were challenged, corrected, or rejected during the process, particularly regarding cardinality notation and normalization decisions.

Prompts used: Prompts were conversational and iterative, covering topics such as ERD design, DFD structure, Phase 3 interface design, SQL schema creation, and Flask web development.